SESSION - 8

1. Pick any one feature from a popular app you use daily (for example: Instagram signup, Flipkart add-to-cart, or Paytm recharge) and write 5 clear requirements for that feature.

Application: Instagram Signup Feature

Requirement Type: Functional Requirements

5 Clear Requirements:

- The user must be able to sign up using a valid email address or mobile phone number.
- The system must require the user to enter a unique username that is not already in use.
- The user must create a password with a minimum of 8 characters.
- The system must display an error message if any mandatory field is left empty or contains invalid data.
- After successful registration, the system must send a verification code to the user's email address or phone number for account verification.

2. For the requirements you wrote in Task 1, create a table mapping each requirement to at least one test case that would verify it works as intended. (Hint: Your table should have columns: Requirement ID, Requirement Description, Test Case ID, Test Case Description.)

Requirements to Test Case Mapping

Req ID	Requirement Description	TC ID	Test Case Description
R1	User must be able to sign up using a valid email address or mobile phone number.	TC01	Enter a valid email address and complete signup. Verify account is created successfully.
R2	User must enter a unique username that is not already in use.	TC02	Enter an existing username and verify the system displays a "Username already exists" message.
R3	User must create a password with a minimum of 8 characters.	TC03	Enter a password with less than 8 characters and verify an error message is displayed.
R4	The system must display an error message if any mandatory field is left empty or contains invalid data.	TC04	Leave the email field empty and click Sign Up. Verify an appropriate error message is shown.
R5	The system must send a verification code to the user's email address or phone number after successful registration.	TC05	Complete signup with valid details and verify that a verification code is sent to the registered email/phone number.

3. Extend your table from Task 2 by adding a 'Defect ID' column. Imagine that during testing, you find 2 bugs related to your test cases. Link these defects to the relevant test cases in your table. (Hint: Defects can be things like 'Signup button not enabled' or 'OTP not received'.)

Test Cases Execution & Defect Mapping

TC ID	Requirement	Test Case Description	Expected Result	Defect ID
TC_01	User can enter a valid mobile number/email	Enter a valid mobile number or email and proceed	User moves to the next step successfully	-
TC_02	User must receive OTP for verification	Enter a valid mobile number and request OTP	OTP should be received within a few seconds	**DEF_002**
TC_03	User can enter a valid username	Enter a unique username	Username is accepted	-
TC_04	User can create a password	Enter a valid password meeting requirements	Password is accepted	-
TC_05	Signup button should be enabled when all mandatory fields are filled	Fill all required fields correctly	Signup button becomes enabled	**DEF_001**
TC_06	User account should be created after successful verification	Complete signup process with valid details and OTP	Account is created successfully	-

Defect Logs

Defect ID	Defect Description	Related Test Case
DEF_001	Signup button remains disabled even after all mandatory fields are filled correctly.	TC_05
DEF_002	OTP is not received on the registered mobile number after requesting verification.	TC_02

4. Explain in 3-4 sentences the difference between forward traceability and backward traceability in the context of your table. Give one example of each using your requirements and test cases.

Forward Traceability ensures that every requirement has at least one corresponding test case to verify complete testing coverage. For example, the requirement *"User must receive OTP for verification"* is linked forward to **TC_02**, which checks if the OTP is received. Conversely, **Backward Traceability** maps test cases back to the parent requirement to verify that no unnecessary tests are created. For instance, **TC_05** *(Signup button should be enabled)* traces backward to the mandatory fields completion requirement. In summary, **Forward Traceability is Requirement → Test Case**, while **Backward Traceability is Test Case → Requirement**.

5. Create a bi-directional RTM (Requirements Traceability Matrix) for the login feature of a food delivery app like Zomato, showing how each requirement is linked to test cases and how test cases trace back to requirements. (Constraint: Include at least 3 requirements and 5 test cases in your matrix.)

Bi-Directional Requirements Traceability Matrix (RTM) – Zomato Login Feature

Req ID	Requirement Description	Test Case ID	Test Case Description
RQ_01	User should be able to log in with a valid mobile number and password.	TC_01	Verify login with valid mobile number and password.
RQ_01	User should be able to log in with a valid mobile number and password.	TC_02	Verify login fails with incorrect password.
RQ_02	User should receive an OTP for mobile number verification.	TC_03	Verify OTP is received on a valid mobile number.
RQ_02	User should receive an OTP for mobile number verification.	TC_04	Verify error message is shown for an invalid OTP.
RQ_03	User should see an error message for invalid login credentials.	TC_02	Verify login fails with incorrect password.
RQ_03	User should see an error message for invalid login credentials.	TC_05	Verify error message is displayed for an unregistered mobile number.

Forward Traceability View (Requirement → Linked Test Cases)

Requirement ID	Linked Test Cases
RQ_01	TC_01, TC_02
RQ_02	TC_03, TC_04
RQ_03	TC_02, TC_05

Backward Traceability View (Test Case → Linked Requirements)

Test Case ID	Linked Requirements
TC_01	RQ_01
TC_02	RQ_01, RQ_03
TC_03	RQ_02
TC_04	RQ_02
TC_05	RQ_03

Note: This bi-directional matrix ensures 100% test coverage for the specified scope, preventing any missing requirements or untraceable test scenarios.